

Investments



- 1 Dave's investment of \$6000 earns interest at 2% p.a, compounded annually over 3 years.
Using repeated multiplication, find:
 - a The value of the investment after 3 years.
 - b The amount of interest earned.
- 2 A \$2090 investment earns interest at 4.2% p.a. compounded annually over 17 years. Use the compound interest formula to calculate the value of this investment.
- 3 Sally's investment of \$8950 earns interest at 4% p.a. compounded annually over 4 years. Find the amount of interest earned.
- 4 \$3900 is invested for three years at a rate of 10% p.a., compounding annually.
 - a Complete the table below to determine the final value of the investment:

	Balance + interest	Total balance	Interest earned
First year	—	\$3900	\$390
Second year	\$3900 + \$390	\$4290	\$429
Third year			
Fourth year			—

- b Calculate the total interest earned over the three years.
- 5 State whether the following is true about compound interest:
 - a Interest is earned on the principal.
 - b The interest in any time period is calculated using only the original principal.
 - c Interest is earned on any accumulated interest.
 - d The amount of interest earned in any time period changes from one period to the next.

Inflation

- 6 What is the value of a \$1330 brand new desk after 7 years when the annual inflation rate is 11%?
- 7 In 2001, the price of a bottle of orange juice was \$3.20. What was the price of a bottle of

- 8 For an inflation rate of 7.4% per year, what is the expected value of a \$210 basket of groceries 7 years from now?
- 9 A one year sports club membership currently costs \$332. Calculate the cost in 6 years time if the inflation rate is 2.6% per annum. Round your answer to the nearest dollar.
- 10 In terms of today's dollar, what will be the value of \$5 in 15 years time if the inflation rate is:

a 2.1%b 4.1%
- 11 Find the original price of the following items:

a A hat today costs \$35.50. Find the price of the hat 9 years ago at an inflation rate of 3.7% per annum.

b An item costs \$15 000 today. Find the price of the item 4 years ago at an inflation rate of 3.5% per annum.
- 12 The price of pens and pencils in 1996 was \$5. If the value inflated at an average rate of 3.2% per annum, what would the price have been in 2005?

Appreciation

- 13 What is the value of a wine collection, currently valued at \$4790, after 10 years if its rate of appreciation is 9% p.a.?
- 14 The value of land is expected to grow by 6.8% each year for the next 5 years. If it currently costs \$460 per square metre, how much will it cost at the end of 5 years?
- 15 Due to dwindling supplies of oil, oil prices are expected to appreciate by 7.2% p.a. If the current price of oil is \$79 a barrel, what is the expected price of a barrel in 23 years time?
- 16 A house was valued 6 years ago to be worth \$548 000. Its value appreciated at 5.2% p.a. What is its appreciated value? Give your answer to the nearest dollar.
- 17 If a piece of land appreciates at an average rate of 3.7% per annum and its current value is \$430 000, calculate its value in 3 years. Give your answer to the nearest dollar.
- 18 A high interest bank account deposit appreciates at 6.6% per annum and is currently valued at \$46 000. What will its value be in 8 years? Give your answer to the nearest dollar.

- 19 Sean deposits \$4000 into a new savings account. The amount increased by 3.3% each year for 4 years. Find the new value of the deposit after appreciation, giving your answer to the nearest dollar.
- 20 The federal government wants to increase funding to a certain program over 8 years at a constant rate of 4.75% per annum over that time period. If the initial funding was \$14 000 000, find the value of the funding (to the nearest dollar) after:
- a 1 year b 4 years c 8 years
- 21 A vintage collectors item that costs \$6000, appreciates at approximately 6.6% p.a. After how many full years, n , will the value of the vintage collectors item be over \$15 000?

Depreciation

- 22 There were 10 100 digital sales of a particular song in its first month of release. The number of monthly digital sales decreased by 360 each month after.
- a The song registered 7580 digital sales in a particular month. What was the decrease in the monthly sales from the first month?
- b In how many months after the initial month of release did the song register 7580 in monthly digital sales?
- 23 Production robots to be used in a car manufacturing plant were purchased for \$4 455 000. After 5 years, they depreciated to a value of \$4 385 000.
- a What was the annual depreciation using the straight-line method?
- b After 7 years, the robots are sold off. If they continue to depreciate at an annual rate of \$14 000, how much can they be sold for?
- 24 The following graph shows the depreciation of a car's value over 4 years:
- a What is the initial value of the car?
- b By how much did the car depreciate each year?
- c After how many years will the car be worth \$14 400?
- d What is the value of the car after 4 years?





25 Marge purchased shares at a total value of  100 000. After 6 months, the shares had dropped in value to \$99 200.

- a** What is the monthly depreciation using the straight-line method?
- b** Sketch the graph showing the depreciation of the value of the shares over 6 years.
- c** What is the gradient of the line?
- d** What does the gradient represent in context?
- e** What is the value of the y -intercept of the line?
- f** What does the y -intercept represent in context?
- g** Let V represent the value of the shares after n months. Write the equation of the line.
- h** Find the value of the shares after $5\frac{1}{2}$ years.